EX-02-Cross-Platform-Prompting-Evaluating-Diverse-Techniques-in-AI-Powered-Text-Summarization

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Subject: Prompt Engineering

Prompt:

The Basics of Blockchain Technology:

Blockchain technology is a decentralized digital ledger system used to record transactions securely and transparently across a network of computers. Unlike traditional databases that rely on a central authority, blockchain operates through a distributed network where each participant maintains a copy of the ledger. This structure reduces the need for intermediaries and increases trust among users. The main idea behind blockchain is to create a system where data can be recorded in a secure, tamper-resistant, and verifiable way. A blockchain is composed of a sequence of blocks that store transaction data. Each block contains three main components: the transaction data, a timestamp, and a cryptographic hash of the previous block. The hash acts like a digital fingerprint, uniquely identifying each block. Because every block contains the hash of the previous block, the blocks are linked together in chronological order, forming a chain. If someone attempts to alter the data in a block, the hash changes, breaking the chain and making the tampering easily detectable. One of the key features of blockchain technology is decentralization. In a decentralized network, control is distributed among many nodes (computers) rather than a single central authority. Each node participates in validating and storing transactions. This structure improves system reliability because the network can continue functioning even if some nodes fail or are attacked. It also increases transparency since all participants can view and verify the transactions recorded on the ledger. To maintain accuracy and trust in the network, blockchain uses consensus mechanisms. These are protocols that allow network participants to agree on the validity of transactions before they are added to the blockchain. Two common consensus mechanisms are Proof of Work (PoW) and Proof of Stake (PoS). In Proof of Work, participants called miners solve complex mathematical puzzles to validate transactions and create new blocks. In Proof of Stake, validators are chosen based on the amount of cryptocurrency they hold and are willing to "stake" as collateral. These mechanisms help prevent fraudulent activities such as double spending. Blockchain technology is widely known for its role in supporting cryptocurrencies like Bitcoin and Ethereum. In

cryptocurrency systems, blockchain enables secure peerto peer transactions without the need for banks or financial intermediaries. Every transaction is recorded on the blockchain and can be verified by anyone on the network, ensuring transparency and accountability. Beyond cryptocurrencies, blockchain has many applications across different industries. In supply chain management, blockchain can track products from their origin to the final consumer, improving traceability and reducing fraud. In healthcare, it can securely store patient records and allow authorized professionals to access them when needed. In digital identity systems, blockchain can help individuals control and verify their personal information without relying on centralized authorities. Despite its advantages, blockchain technology also faces challenges such as high energy consumption in some consensus mechanisms, scalability limitations, and regulatory concerns. However, ongoing research and technological advancements aim to address these issues and improve the efficiency of blockchain systems. In conclusion, blockchain technology provides a secure, transparent, and decentralized method of recording and verifying transactions. Its unique structure and consensus mechanisms make it a powerful tool for building trust in digital systems, and its potential applications continue to expand across various sectors. summarize this paragraph.

Output:

Summary of Blockchain Technology

Blockchain technology is a decentralized digital ledger that records transactions across a network of computers in a secure, transparent, and tamper-resistant way. Instead of relying on a central authority such as a bank or government, blockchain operates through a distributed network where each participant maintains a copy of the ledger. Transactions are grouped into blocks, and each block is linked to the previous one using cryptographic techniques, forming a continuous chain. This structure ensures that once information is recorded, it is extremely difficult to alter or delete.

One of the key features of blockchain is transparency. All participants in the network can view the recorded transactions, which helps build trust among users. Security is maintained through cryptography and consensus mechanisms, which are methods used by network participants to agree on the validity of transactions before they are added to the blockchain. Common consensus methods include Proof of Work (PoW) and Proof of Stake.

Blockchain technology is widely known for powering cryptocurrencies such as Bitcoin, but its applications extend beyond digital currency. It can be used in areas such as supply chain management, healthcare, finance, voting systems, and digital identity verification. By providing a secure and transparent way to record and verify data,

blockchain reduces the need for intermediaries, lowers costs, and increases efficiency in many processes.

Despite its advantages, blockchain also faces challenges such as scalability, high energy consumption in some consensus models, and regulatory concerns. However, ongoing research and development aim to improve the technology and expand its practical applications. Overall, blockchain has the potential to transform how data and transactions are managed across various industries.

GEMINI:

Blockchain is a decentralized, digital ledger technology that records transactions across a network of computers in a way that is secure, transparent, and permanent. Unlike traditional databases managed by a central authority (like a bank), blockchain is maintained by a distributed network of participants, making it resistant to tampering and fraud.

Core Principles

- **Decentralization:** No single entity controls the data. Instead, the ledger is shared among all participants (nodes) in the network, ensuring that everyone has an identical copy of the record.
- **Immutability:** Once a transaction is verified and added to the "chain" in a data "block," it cannot be altered or deleted. Each block is cryptographically linked to the one before it, meaning any change would break the entire chain.
- **Consensus:** For a new block to be added, the network must agree on its validity through specific protocols (like Proof of Work or Proof of Stake), eliminating the need for a trusted middleman.
- **Cryptography:** Heavy encryption protects the data and identifies users via public and private keys, ensuring privacy while maintaining a public audit trail.

Key Benefits

- **Transparency:** All participants can view the transaction history, which builds trust and accountability.
- **Security:** The lack of a central point of failure makes the network highly resilient against hacking and unauthorized access.
- **Efficiency:** By removing intermediaries and automating processes with **Smart Contracts** (self-executing code), blockchain reduces costs and speeds up settlement times.

- **Traceability:** It provides an end-to-end audit trail, which is particularly valuable for verifying the origin of goods in a supply chain.

Common Applications

Beyond its most famous use in **cryptocurrencies** like Bitcoin, blockchain is utilized in various sectors:

- **Finance:** For faster cross-border payments and decentralized finance (DeFi).
- **Supply Chain:** To track products from raw materials to the end consumer.
- **Healthcare:** To securely manage and share patient medical records.
- **Real Estate:** To automate property transfers and digitize land titles.

COPILOT:

Structure: Data is stored in blocks, each containing transactions, a timestamp, and a cryptographic hash. Blocks are linked together, forming an immutable chain.

- **Decentralization:** Unlike traditional databases controlled by a single entity, blockchain distributes data across multiple computers (nodes).
- **Consensus Mechanism:** Transactions are validated by majority agreement (e.g., Proof of Work, Proof of Stake), ensuring trust without intermediaries.
- **Immutability:** Once recorded, data cannot be altered without breaking the chain, making tampering easily detectable.
- **Transparency:** All participants can view the ledger, fostering accountability and trust.

Applications Across Industries

- **Finance:** Enables cryptocurrencies like Bitcoin and Ethereum, cross-border payments, and decentralized finance (DeFi).
- **Supply Chain:** Tracks goods from origin to consumer, ensuring authenticity and reducing fraud.
- **Healthcare:** Secures patient records and enables safe data sharing.
- **Governance:** Supports transparent voting systems and public record management.
- **Smart Contracts:** Self-executing agreements coded directly into the blockchain, reducing reliance on intermediaries.

Challenges & Limitations

- **Energy Consumption:** Proof of Work systems (like Bitcoin) require massive computational power.
- **Scalability Issues:** High transaction volumes can slow networks.
- **Regulatory Uncertainty:** Governments worldwide are still defining policies around blockchain and cryptocurrencies.
- **Integration Barriers:** Legacy systems may struggle to adopt blockchain **solutions.**

Summary

Blockchain is **not just about cryptocurrencies**—it's a foundational technology reshaping how data, assets, and trust are managed globally. Its **decentralized, immutable, and transparent nature** makes it a powerful tool for industries ranging from finance to healthcare, though challenges like scalability and regulation remain.

Would you like me to create a **visual infographic-style summary** of blockchain's structure and applications to make it easier to grasp at a glance?

CLAUDE:

Blockchain is a decentralized, distributed digital ledger that records transactions across a network of computers. Key points include:

- **Structure**: Data is stored in "blocks" that are cryptographically linked in a chronological "chain," making records tamper-resistant and immutable.
- **Decentralization**: No single authority controls the network; consensus mechanisms (like Proof of Work or Proof of Stake) validate transactions collectively.
- **Transparency & Security**: All participants can view the ledger, while cryptographic hashing ensures data integrity and prevents unauthorized alterations.
- **Smart Contracts**: Self-executing code on blockchains (e.g., Ethereum) automates agreements without intermediaries.
- **Applications**: Beyond cryptocurrency (Bitcoin), blockchain is used in supply chain management, healthcare records, voting systems, finance (DeFi), and digital identity verification.

- **Challenges**: Scalability, energy consumption, regulatory uncertainty, and interoperability between different blockchains remain ongoing concerns.

In essence, blockchain provides a secure, transparent, and trustless way to record and verify data, with the potential to disrupt industries that rely on centralized record-keeping and intermediaries.

Prompt:

The Basics of Blockchain Technology : Blockchain technology is a decentralized digital ledger system used to record transactions securely and transparently across a network of computers. Unlike traditional databases that rely on a central authority, blockchain operates through a distributed network where each participant maintains a copy of the ledger. This structure reduces the need for intermediaries and increases trust among users. The main idea behind blockchain is to create a system where data can be recorded in a secure, tamper-resistant, and verifiable way. A blockchain is composed of a sequence of blocks that store transaction data. Each block contains three main components: the transaction data, a timestamp, and a cryptographic hash of the previous block. The hash acts like a digital fingerprint, uniquely identifying each block. Because every block contains the hash of the previous block, the blocks are linked together in chronological order, forming a chain. If someone attempts to alter the data in a block, the hash changes, breaking the chain and making the tampering easily detectable. One of the key features of blockchain technology is decentralization. In a decentralized network, control is distributed among many nodes (computers) rather than a single central authority. Each node participates in validating and storing transactions. This structure improves system reliability because the network can continue functioning even if some nodes fail or are attacked. It also increases transparency since all participants can view and verify the transactions recorded on the ledger. To maintain accuracy and trust in the network, blockchain uses consensus mechanisms. These are protocols that allow network participants to agree on the validity of transactions before they are added to the blockchain. Two common consensus mechanisms are Proof of Work (PoW) and Proof of Stake (PoS). In Proof of Work, participants called miners solve complex mathematical puzzles to validate transactions and create new blocks. In Proof of Stake, validators are chosen based on the amount of cryptocurrency they hold and are willing to “stake” as collateral. These mechanisms help prevent fraudulent activities such as double spending. Blockchain technology is widely known for its role in supporting cryptocurrencies like Bitcoin and Ethereum. In cryptocurrency systems, blockchain enables secure peerto peer transactions without the need for banks or financial intermediaries. Every transaction is

recorded on the blockchain and can be verified by anyone on the network, ensuring transparency and accountability. Beyond cryptocurrencies, blockchain has many applications across different industries. In supply chain management, blockchain can track products from their origin to the final consumer, improving traceability and reducing fraud. In healthcare, it can securely store patient records and allow authorized professionals to access them when needed. In digital identity systems, blockchain can help individuals control and verify their personal information without relying on centralized authorities. Despite its advantages, blockchain technology also faces challenges such as high energy consumption in some consensus mechanisms, scalability limitations, and regulatory concerns. However, ongoing research and technological advancements aim to address these issues and improve the efficiency of blockchain systems. In conclusion, blockchain technology provides a secure, transparent, and decentralized method of recording and verifying transactions. Its unique structure and consensus mechanisms make it a powerful tool for building trust in digital systems, and its potential applications continue to expand across various sectors. summarize this paragraph in Zero-Shot ,Few-Shot ,Chain-of-Thought ,Role-Based Prompt Technique

Output:

GEMINI:

Blockchain is a decentralized, distributed ledger technology that ensures data integrity through a network of peer-to-peer nodes rather than a central authority. Its architecture relies on **immutability**, where cryptographic hashing links sequential blocks, making retrospective alteration computationally infeasible. Transactions are validated via **consensus protocols**, which eliminate intermediaries while ensuring network-wide synchronization. By utilizing public-key cryptography and **smart contracts**, blockchain facilitates transparent, secure, and automated processes across diverse sectors, including finance and supply chain management. This shift from trust-based to verification-based systems minimizes fraud and operational costs. Ultimately, blockchain represents a fundamental paradigm shift in digital trust, providing a robust framework for a more transparent, secure, and decentralized global economy.

CHATGPT:

Blockchain technology is a decentralized digital ledger that records transactions across a distributed network of computers, eliminating the need for a central authority. Transactions are grouped into blocks and linked using cryptographic techniques, forming an immutable chain that ensures data integrity and security. The system relies on consensus mechanisms such as Proof of Work and Proof of Stake to validate and add transactions to the network. Blockchain provides transparency, as all participants can view verified records, fostering trust among users. Although widely recognized for

enabling cryptocurrencies, its applications extend to supply chains, healthcare, finance, voting systems, and digital identity management. Despite challenges like scalability and energy consumption, blockchain remains a transformative technology with the potential to fundamentally reshape secure data management and digital transactions across industries.

COPILOT:

Blockchain technology is a decentralized digital ledger that securely records transactions across a distributed network of computers. It organizes data into blocks linked by cryptographic hashes, ensuring immutability and transparency. Consensus mechanisms such as Proof of Work and Proof of Stake validate transactions without central authorities, fostering trust among participants. Its applications extend beyond cryptocurrencies to finance, supply chain management, healthcare, governance, and smart contracts, offering efficiency, security, and accountability. Despite challenges like energy consumption, scalability, and regulatory uncertainty, blockchain represents a transformative innovation. By enabling tamper-resistant, transparent, and trustless systems, it holds profound significance as a foundational technology reshaping global data management and digital interactions.

CLAUDE:

Blockchain is a decentralized, distributed ledger that records transactions across a computer network in cryptographically linked blocks, ensuring immutability and tamper resistance. Governed by consensus mechanisms such as Proof of Work or Proof of Stake, it eliminates the need for central authority. Its transparent architecture allows all participants to verify data integrity, while smart contracts enable automated, intermediary-free agreements. Applications span cryptocurrency, supply chain, healthcare, finance, and digital identity. However, challenges including scalability, energy consumption, and regulatory uncertainty continue to limit widespread adoption.

Blockchain represents a foundational shift in how trust and data integrity are established, positioning itself as a transformative infrastructure for the decentralized digital economy.

GEMINI:

Analysis of Components

- **Main Thesis:** Blockchain is a decentralized, distributed ledger technology that establishes a secure, transparent, and permanent record of transactions without the need for a central authority.

- **Supporting Arguments:** The system's integrity is maintained through four pillars: **decentralization** (shared control), **immutability** (permanent records via cryptographic linking), **consensus protocols** (collective validation), and **cryptography** (data protection). These features result in increased operational efficiency, enhanced security against fraud, and superior traceability across various industries.
- **Conclusion:** Beyond its origins in cryptocurrency, blockchain serves as a versatile tool for sectors like finance, healthcare, and supply chain management by replacing traditional trust-based intermediaries with automated, verifiable systems.

Executive Summary

Blockchain technology functions as a decentralized digital ledger that eliminates the need for central intermediaries by distributing record-keeping across a global network. By integrating **cryptographic immutability** and **consensus protocols**, the architecture ensures that data remains tamper-proof and universally verifiable. This framework provides significant advantages in transparency, security, and efficiency—notably through the use of **smart contracts** to automate complex processes.

While initially popularized by cryptocurrencies, the technology's ability to provide an immutable audit trail has led to its adoption in finance, healthcare, and supply chain logistics. Ultimately, blockchain represents a foundational shift in data management, moving from centralized silos to a transparent, verification-based infrastructure.

CLAUDE:

Analytical Breakdown

Main Thesis Blockchain is a decentralized digital ledger that enables secure, transparent, and recording of transactions without reliance on a central authority.

Supporting Arguments

- Cryptographically linked blocks ensure data is immutable and tamper-resistant
- Consensus mechanisms (Proof of Work/Stake) collectively validate transactions, removing single-point control
- Cryptographic hashing guarantees transparency and data integrity across all participants

- Smart contracts automate agreements, eliminating intermediaries on platforms like Ethereum
- Real-world applications extend across healthcare, supply chain, finance, voting, and digital identity

Conclusion Despite challenges in scalability, energy use, regulation, and interoperability, blockchain holds transformative disruptive potential for centralized industries.

Final Summary

Blockchain is a decentralized, distributed ledger that records transactions in cryptographically linked, immutable blocks — validated collectively through consensus mechanisms, requiring no central authority. Its transparent architecture ensures data integrity, while smart contracts automate agreements without intermediaries. Applications reach well beyond cryptocurrency into healthcare, supply chain, decentralized finance, and digital identity. Though challenges such as scalability, energy consumption, and regulatory ambiguity persist, blockchain's core innovation — establishing trust through cryptography rather than institutions — positions it as a foundational technology capable of fundamentally restructuring how data, value, and agreements are managed across industries.

COPILOT:

Here’s a clear breakdown of the 500-word passage into its key components, followed by a concise summary:

Key Components

Main Thesis Blockchain is a transformative digital ledger technology that ensures secure, transparent, and decentralized record-keeping, with wide-ranging applications beyond cryptocurrency.

Supporting Arguments

• Structure & Integrity: Blockchain is a chain of blocks linked by cryptographic hashes, making records immutable and tamper-resistant.

• Peer-to-Peer Network: Transactions are verified by multiple nodes without a central authority, using consensus mechanisms.

• Consensus Algorithms: o Proof of Work (PoW): Secure but energy-intensive. o Proof of Stake (PoS): More energy-efficient, based on validators’ staked assets.

• Decentralization & Transparency: Eliminates single points of failure, distributes control, and ensures visibility of transaction history.

• Smart Contracts: Self-executing programs that automate agreements, enabling decentralized applications (dApps).

• Challenges: Scalability, energy consumption, regulatory uncertainty, and interoperability issues hinder adoption.

Conclusion: Blockchain redefines digital trust by combining cryptography, decentralization, and transparency. Despite challenges, ongoing innovation positions it as a foundational technology for the future digital economy.

CHATGPT:

Key Components Breakdown Main Thesis:

Blockchain technology is a decentralized, secure, and transparent distributed ledger system that enables tamper-resistant record-keeping without relying on a central authority.

Supporting Arguments:

1. Structure and Security: Blockchain consists of linked blocks containing transaction data and cryptographic hashes, ensuring immutability and data integrity.

2. Decentralization and P2P Network: Multiple nodes maintain and verify the ledger, eliminating single points of failure.

3. Consensus Mechanisms: Algorithms like Proof of Work (PoW) and Proof of Stake (PoS) validate transactions and secure the network.

4. Transparency and Trust: Public blockchains allow visible transaction histories while maintaining pseudonymity.

5. Smart Contracts: Self-executing programs automate agreements and enable decentralized applications.

6. Challenges: Scalability, energy consumption, regulatory uncertainty, and interoperability limit adoption. Conclusion: Blockchain represents a transformative digital trust system with growing potential despite existing technical and regulatory challenges.

Summary

Blockchain technology is a decentralized distributed ledger system that ensures secure, transparent, and tamper-resistant record-keeping. It functions through a chain

of cryptographically linked blocks, where each block stores verified transaction data. Operating on a peer-to-peer network, blockchain removes the need for central authorities by relying on consensus mechanisms such as Proof of Work and Proof of Stake to validate transactions. Transparency and decentralization enhance trust and resilience, while smart contracts expand functionality by automating agreements. Although challenges like scalability, energy usage, and regulatory uncertainty persist, blockchain continues to evolve as a foundational technology for the future digital economy.

ChatGPT:

1.Zero-Shot Prompting Summary:

Blockchain is a decentralized digital ledger that records transactions across a network of computers without a central authority. Transactions are stored in blocks linked by cryptographic hashes, making the data secure and tamper-resistant. The network verifies transactions using consensus mechanisms like Proof of Work (PoW) and Proof of Stake (PoS). Blockchain is widely used in cryptocurrencies such as Bitcoin and Ethereum and has applications in supply chains, healthcare, and digital identity systems. Despite advantages like transparency and security, it faces challenges such as scalability, energy consumption, and regulatory issues.

2. Few-Shot Prompting Summary:

Blockchain technology is a distributed and decentralized system that securely records transactions across multiple computers. Data is stored in blocks connected through cryptographic hashes, forming a tamperresistant chain. Transactions are verified through consensus methods such as PoW and PoS, ensuring trust without a central authority. While blockchain powers cryptocurrencies like Bitcoin and Ethereum, it is also used in sectors like healthcare, supply chain management, and digital identity. The technology improves transparency and security but still faces challenges related to scalability and energy usage.

3. Chain-of-Thought Prompting

Step-by-Step Reasoning Summary:

1. Definition: Blockchain is a decentralized ledger used to record and verify transactions across a distributed network.

2. Structure: Transactions are grouped into blocks containing data, timestamps, and cryptographic hashes linking them together.

3. Security: The linking of blocks ensures that any data modification is easily detected.

4. Validation: Consensus mechanisms such as Proof of Work and Proof of Stake verify transactions before adding them to the chain.

5. Applications: Blockchain powers cryptocurrencies like Bitcoin and Ethereum and is also used in industries like healthcare and supply chains.

6. Conclusion: While blockchain offers transparency, security, and decentralization, it still faces challenges like scalability, energy consumption, and regulatory concerns.

4. Role-Based Prompting Role:

Technology Analyst As a technology analyst, the article explains that blockchain technology is a decentralized system for securely recording transactions across multiple computers. It uses cryptographic hashing and distributed consensus mechanisms to ensure data integrity and transparency without relying on a central authority. The technology is most recognized for enabling cryptocurrencies such as Bitcoin and Ethereum. Beyond digital currencies, blockchain supports applications in supply chain tracking, healthcare data management, and digital identity verification. Although it provides strong security and trust, ongoing improvements are needed to address issues like scalability, energy usage, and regulatory frameworks

Gemini:

1.Zero-Shot

Directly summarizing the content without prior examples.

Summary: Blockchain is a decentralized digital ledger that records transactions across a network of computers using linked "blocks" secured by cryptographic hashes. By utilizing consensus mechanisms like Proof of Work and Proof of Stake, it ensures security and transparency without a central authority. While it famously powers cryptocurrencies, its applications extend to supply chains and healthcare, though it still faces hurdles like scalability and energy use.

2. Few-Shot

Simulating a pattern based on examples (e.g., Topic: One-sentence summary).

Input Example 1: [Article about AI]-> AI mimics human intelligence to solve complex tasks.

Input Example 2: [Article about Solar Power]-> Solar power converts sunlight into electricity for sustainable energy.

Summary: Blockchain technology provides a secure, decentralized way to verify and store data across a global network.

3. Chain-of-Thought

Breaking down the logic step-by-step before reaching a conclusion.

• Step1(Core Identity): It is a decentralized, distributed ledger system.

• Step2(Mechanism): Blocks are linked by hashes, making them tamperresistant.

• Step3(Validation): Consensus protocols (PoW/PoS) verify data without intermediaries

. • Step4(Utility & Challenges): It is used in crypto and industry but faces energy and scaling issues.

Summary:

Consequently, blockchain is a secure framework that uses cryptographic linking and collective validation to create a transparent, multiindustry tool for trustless data management.

4.Role-Based Role:

Tech Consultant for Beginners "Think of blockchain as a shared digital record book that nobody can cheat. Instead of one bank keeping the books, everyone in the network has a copy. It uses clever math (hashes) to lock information in a chain, making it nearly impossible to hack. While it’s the engine behind Bitcoin, it’s also revolutionizing how we track products and protect medical data, despite some current 'growing pains' regarding speed and power usage."

Claude:

1.Zero-Shot Prompt

Prompt: "Summarize the article 'The Basics of Blockchain Technology' in 100 words or less, covering definition, structure, features, and applications."

Summary: Blockchain is a decentralized ledger recording transactions in linked blocks via cryptographic hashes for tamper-resistance. Key features: decentralization across nodes, consensus like PoW (mining puzzles) or PoS (staking). Powers crypto (Bitcoin, Ethereum); applies to supply chains, healthcare, identities. Challenges: energy use, scalability.

2.Few-Shot Prompt

Prompt:

"Example 1:

Input: Brief on AI.

Output: AI mimics human intelligence via machines learning from data.

Example 2:

Input: What is cloud computing?

Output:

Cloud computing delivers on-demand resources over internet, scalable without owning hardware.

Summary: Blockchain is a decentralized ledger of transaction blocks linked by hashes, ensuring security without central control. Nodes validate via PoW or PoS consensus; enables crypto like Bitcoin, plus supply chain tracking, health records, identities. Faces energy and scale issues.

3.Chain-of-Thought Prompt

Prompt: "Summarize the article step-by-step:

1. Identify main topic.

2. List key components (structure, features).

3. Note mechanisms and apps.

4. Mention challenges.

5. Combine into 1 paragraph."

Summary:

1. Main topic: Basics of blockchain as decentralized ledger.

2. Structure: Blocks with data, timestamp, prior hash; chain links them. Features: Decentralized nodes for reliability/transparency.

3. Mechanisms: PoW (puzzles), PoS (staking); apps: Crypto (Bitcoin/Ethereum), supply chain, healthcare, IDs.

4. Challenges: Energy, scalability, regs. Combined: Blockchain chains secure transaction blocks across decentralized nodes using PoW/PoS consensus, powering crypto and industries like supply chains despite energy hurdles.

4.Role-Based Prompt

Prompt: "You are a tech educator. Explain 'The Basics of Blockchain Technology' article to a beginner in a 150-word summary, using simple **analogies."**

Summary:

Imagine blockchain as a shared notebook where everyone writes transactions in pages (blocks) locked with unique codes (hashes) linking to the previous page—no single boss in charge. Computers (nodes) copy the notebook for backup and vote on additions via puzzles (PoW) or bets (PoS) to keep it honest. Famous for Bitcoin cash swaps without banks; also tracks food from farm to table, secures health files, and manages IDs. It's super secure and open, but guzzles energy and grows slowly—fixes are coming!

Copilot:

1. Zero-Shot

Summary: Blockchain is a decentralized digital ledger that records transactions securely and transparently across a network of computers. It links blocks of data using cryptographic hashes, making tampering detectable. Consensus mechanisms like Proof of Work and Proof of Stake ensure trust and prevent fraud. While best known for cryptocurrencies, blockchain also supports applications in supply chains, healthcare, and digital identity. Challenges include energy use, scalability, and regulation, but ongoing research seeks solutions.

2. Few-Shot Example

Summaries (few-shot style):

• Example 1: "AI is reshaping industries by automating tasks and improving efficiency."

• Example 2: "Climate change is driven by human activity and requires urgent global action."

Summary (following the examples):

Blockchain is a secure, decentralized ledger system that links blocks of data with cryptographic hashes. It uses consensus mechanisms to validate transactions, supports cryptocurrencies, and has broader applications in supply chains, healthcare, and identity management. Despite challenges like energy consumption and scalability, it remains a transformative technology.

3. Chain-of-Thought

Reasoning:

1. Blockchain is defined as decentralized and secure. 2.
2. Its structure: blocks + hashes + timestamps.
3. Consensus mechanisms ensure trust (PoW, PoS).
4. Applications: cryptocurrencies, supply chain, healthcare, identity.

Summary: Challenges: energy, scalability, regulation. Conclusion: powerful but evolving technology. Blockchain is a decentralized, tamper-resistant ledger that uses cryptographic links and

consensus mechanisms to ensure trust. It underpins cryptocurrencies and offers applications in industries like supply chain and healthcare, though it faces energy and scalability challenges.

5. Role-Based Prompt Technique

• Teacher:

"Blockchain is like a digital notebook shared across many computers. Each page (block) is linked to the previous one with a special code (hash), making it hard to cheat. People agree on what gets written using rules called consensus mechanisms. It's used in Bitcoin but also in healthcare and supply chains."

• Business Analyst:

"Blockchain provides decentralized trust by eliminating intermediaries. Its applications extend beyond cryptocurrency into supply chain transparency, secure health records, and digital identity. Key challenges include scalability and regulatory frameworks."

• Technologist:

"Blockchain is a distributed ledger secured by cryptographic hashes and consensus protocols such as PoW and PoS. It ensures immutability and transparency, enabling peer-to-peer transactions and industry applications, though energy efficiency and scalability remain pressing issues."

Comparison:

Prompt Technique	Platform	Accuracy	Coherence	Simplicity	Speed	UX
ZERO-SHOT	**ChatGPT**	4.2	4.5	4.0	5.0	5.0
	Gemini	**4.9**	**4.8**	**4.5**	**4.8**	4.5
	Copilot	3.0	3.9	4.2	4.1	4.0
FEW-SHOT	**ChatGPT**	4.0	4.3	4.0	5.0	5.0
	Gemini	**5.0**	**4.5**	**4.0**	**4.8**	4.9
	Copilot	3.3	4.0	4.0	4.3	3.0
CHAIN-OF-THOUGHT	**ChatGPT**	4.0	4.7	4.0	5.0	4.0
	Gemini	**4.8**	**4.7**	**4.5**	**4.5**	5.0

Prompt Technique	**Platform**	**Accuracy**	**Coherence**	**Simplicity**	**Speed**	**UX**
	Copilot	4.0	3.9	4.0	4.3	3.0
ROLE-BASED	**ChatGPT**	3.8	4.5	4.0	5.0	4.7
	Gemini	**4.7**	**4.8**	**4.5**	**4.8**	5.0
	Copilot	3.9	4.0	4.0	4.0	4.0